

ISA Exploratorium Summary

The many computer designs of Jim Brakefield

James Brakefield

Mr. Brakefield finds computer architecture fascinating. Historically early computers were accumulator based with economy of design being very important. Memory sizes were small, even tiny, by modern standards. Word size was set by the application in mind. Production runs were limited, machines were manually wired.

The invention of the transistor and core memory led to what might be called the modern era. Production runs increased into the hundreds and thousands. Scientific memory sizes were in the tens of thousands of words which varied from, say, 16 to 60 bits. Indirect addressing and a small number of address size index registers were the norm. Business computers were character based with similar address sizes. Most used modular printed circuit boards plugged into a hand wired backplane.

The arrival and dominance of the IBM 360 led to standardization on byte addressing and 8-bit character sets. Previously each new computer came with its own unique 6-bit character set. Although the 360 was capable of 32-bit addressing, it took time before four gigabyte memory was available. Virtual memory allowed machines with modest memory to support much larger address spaces and the associated much larger software programs.

With the emergence of small scale integrated circuits and solid state memory a variety of 16 to 32-bit word size "laboratory" mini computers emerged. These tended to have register files of eight to 32 registers usable as both index/address registers and accumulators. One of these "mini-computers", the DEC PDP11 series pioneered a most efficient 32-bit floating point. This led to the floating-point standard "IEEE 754" which soon dominated as it was implemented on the microprocessors and became universally supported.

Somewhere on this timeline the single chip microprocessor appeared. Cost and power requirements plummeted heralding the era of personal computers.

At somewhat the same time integrated circuit technology improved to where a single chip 32-bit computer was possible. Economy of design prevailed resulting in uniform register files of 32-bit words and an associated address space. There was a transition to multi-layer circuit boards and surface mounted components.

Now some fifty years later, IC technology has progressed to where inexpensive 32-bit processors are the norm. With modest on chip memory capacity these serve the embedded market. With larger external memory, often of gigabytes, and gigahertz performance, they find their way into personal computers, cell phones and super computers (that have thousands of networked processor chips).

The most recent era consists of lowered cost of custom chip fabrication that allows custom chip design at the university level. The decreasing chip costs and increased performance has made the FPGA implementation of custom processors available to one and all. FPGAs find use in low volume products and as prototypes for high volume products.

This is a very brief history of digital computers. Several better or more complete histories are on the internet. Jim Brakefield participated in this computer revolution and is familiar with several of the IC computer families and their Instruction Set Architecture. He has not lost his desire to be a computer architect, although it is a demanding occupation. The arena for ISA design spans the corporate world, where legacy architecture dominates, the educational world of very modest architectures for course work, the legacy world of emulating various computer systems and various gaming machines, and finally the hobby world that designs and implements ISAs, as the British say, “for the hell of it”.

Mr. Brakefield’s work can be easily classified as “legacy upgrades”. These are ideas for improving several successful microprocessor designs to include more address space, a larger register file, a more efficient ISA all within the context of FPGA implementation. Now there are literally dozens if not hundreds of such efforts to be found on the internet, often at GitHub web sites as open source.

On the educational side of this, previously the MIPS architecture was often used at the university level. Today there are thousands of such projects using the RISC-V architecture. Both of these are 32-bit processors with uniform register files and 32-bit instructions.

So what has Mr. Brakefield done?

They have common characteristics: Variable length instructions, variable length immediate values and utilize a register file. For those that support a 16 or 32 registers, one of the register fields can indicate the size of a following immediate value as well as provide a small number of immediate values.

An ultimate goal is to compare stack and register machines on an equal basis. Also derive LUT costs and performance for various instruction groupings and data types.

The ideas are numerous, the implementations (e.g. working and tested RTL) much less so:

LEM1_9 & LEM4_9 Single bit and four bit accumulator designs: working & tested

Goal is a very small processor suitable for logic emulation and BCD processing.

There are variants with pointer registers, suitable address modes and a return stack

ROC24_24min RISC design with 24-bit register file, 24-bit instructions and addressing, RTL & tested

Goal is a simple single cycle RISC ISA with instructions more compact than 32-bit RISC.

It has a register file of 64 24-bit registers

TROC16 RISC design with 16-bit register file; 16, 24 & 32-bit instructions: ~full RTL

Goal is improved code density over RISC-V and a better ISA

Innovations: four data types and four data sizes, to be fully implemented on TROC24

A register “residue”, part of the register file, captures additional information from ALU instructions that normally would go to a status register or would not be captured at all.

Mr. Brakefield's ISA Playground (not completely implemented)

BCD	Variations on IBM1620 having variable length (flag bit delimited) data fields	
AltBCD	non-BCD digits used for sign, field end flag, record flag	
Base100	7-bit digit plus flag-bit per byte: radix 100 and radix 128	
	Quantitative comparisons on data and program size:	
	IBM1620, AltBCD, Base100, IBM360/B2500 and today's uP data formats	
LEM	Logic Emulation Machine	
Lem1_9min	single bit accumulator, 9 & 18 bit instructions	
Lem1_9ptr	Four pointers with four address modes, return stack	
Lem4_9ptr	Four bit accumulator, binary & BCD add/sub, ibid.	
ROIS	Register Oriented Instruction Set	
Rois24_24min	24-bit ALU & memory, word only	
Rois24_3sz	Three data sizes (8, 16,24) and three data types (unsigned, signed, float)	
TROC	Tagged register oriented computer	
Troc16_16min	first version using 16-bit instructions & 16-bit immediate	
Troc16	current version, dual port register file, half-word aligned	
Troc16_4234	ultimate 16-bit version, 24-bit instructions for load/store	
Troc24_8234	full TROC implementation, 24-bit ALU, 8-bits of tag	
OOSM	One operand stack machine, accumulator/stack oriented computer	
Tasoc18_18	Tagged stack values, stacks mapped to register file	
Astoc11_16	8, 9 & 10-bit byte encodings	
Quad_isa	Unification of stack, register and register-memory ISAs with macro facility	
UNA8U_8XS	stack, accumulator, risc & x86 like ISAs	
Legacy uP upgrades	provide more data sizes, types or registers for legacy uP	
	Surprisingly all have 16 and 24-bit instructions, byte alignment and variable length immediate values	
Alt-780	Alternative VAX-780: one data memory reference per instruction, simplified addressing modes	
Alt-11	Alternative PDP-11, five address modes and 12 registers + PC & immediate values	
	DSO8U_16X_def... at _jcb_uP; see uP_log 09/21/2021, 06/11/2021, 01/18/2021	
Alt-430	Alternative MSP-430, five address modes and 14 larger registers	
Alt-x86	Alternative x86, no prefix bytes, uniform instruction formats, 16 registers	
Others	abandoned	
HayStk	Quad issue RIOS (four ALUs)	has RTL & ISE project
LEM16_18	RISC with bit field instructions	has RTL & ISE project
The_12X_uP	Stack machine with 12/24-bit instructions	has RTL & ISE project
CnSIS	Combined "C" oriented and dual stack instruction set	2013 docs
MOIS8U_16	Memory oriented instruction set	2009 docs
ACC18_18abs	Accumulator based architecture, alt-1604?	has RTL & ISE project
UNA8U_8XS	Bit adrs, variable length op-code, operand & result fields	2009 docs

More TROC

As a tribute to 36 and 48-bit processors of the past, ideas are presented on how to efficiently support 36, 42 and 48-bit word sizes on FPGA or ASIC using standard DRAM. It is part of the TROC architecture wherein the instructions occupy 8-bit byte memory and data has a “byte size” of 9, 21 or 12-bits.

TROC with its four data types can support the Radix100 using register pointers as the fourth data type.